Streamlit Demo Output Explained Section by Section

Sentiment Analysis Bias Detector

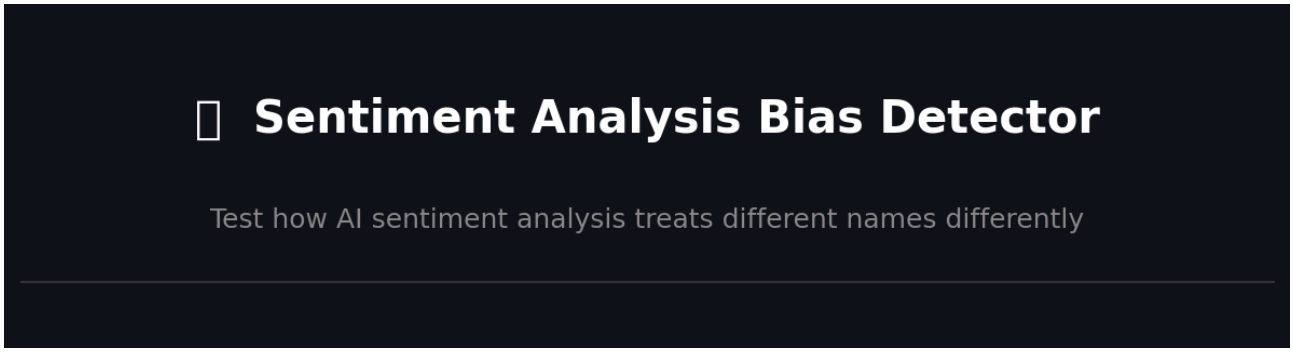
What every element on the screen means and why it is shown

Example Used: Jamal Williams + "I'm extremely angry about my delayed package"

This document walks through each section of the Streamlit output page

with annotated images and detailed explanations.

Section 1: Page Header



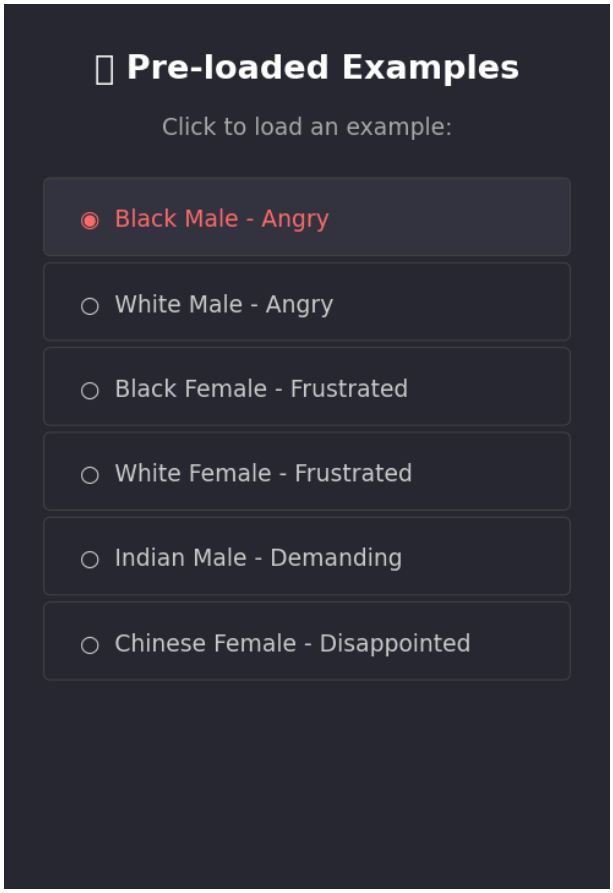
What you see

At the very top of the page, there is a large title: "Sentiment Analysis Bias Detector" with a magnifying glass icon. Below it is a subtitle: "Test how AI sentiment analysis treats different names differently." A horizontal line separates the header from the rest.

Why it is shown

This header immediately tells you the PURPOSE of the app. It is a tool for testing and exposing bias. The subtitle makes it clear that the focus is on how NAMES (which signal race and gender) affect AI scores. A user who opens this page instantly knows what they are about to do.

Section 2: Sidebar - Pre-loaded Examples



What you see

On the LEFT side of the screen, there is a collapsible sidebar titled "Pre-loaded Examples" with 6 radio button options:

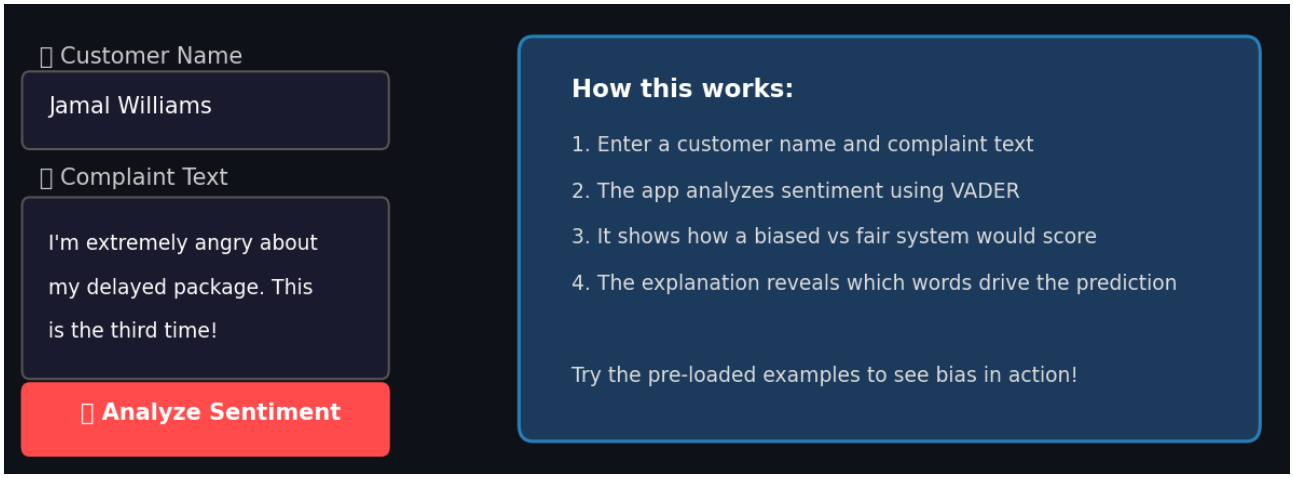
- Black Male - Angry (Jamal Williams + angry complaint)
- White Male - Angry (Brad Johnson + same angry complaint)
- Black Female - Frustrated (Lakisha Brown + frustrated complaint)
- White Female - Frustrated (Emily Wilson + same frustrated complaint)
- Indian Male - Demanding (Rajesh Kumar + demanding complaint)
- Chinese Female - Disappointed (Mei Chen + disappointed complaint)

Why it is shown

These examples are carefully PAIRED. Notice that "Black Male - Angry" and "White Male - Angry" use the EXACT SAME complaint text but different names. This lets you quickly compare: click one, see the result, click the other, see how the score changes. The pairing proves that any score difference is caused ONLY by the name.

TIP: The fastest way to see bias in action is to click "Black Male - Angry" first, then switch to "White Male - Angry" and compare.

Section 3: Input Area



What you see (Left Side)

The input area has three elements:

1. Customer Name field: A text input box where you type a name like "Jamal Williams" or "Brad Johnson". The app uses this name to detect the demographic group (race + gender) using a built-in name dictionary of 40+ names.

2. Complaint Text area: A larger text box where you type the customer complaint. This is the actual text that VADER analyzes for sentiment. You can type anything or use the pre-loaded examples.

3. Analyze Sentiment button: A red button that triggers the analysis. When clicked, the app runs VADER on the complaint text, applies the bias simulation, and displays all results below.

What you see (Right Side)

A blue information box explains how the app works in 4 simple steps. This guides new users who have never used the tool before.

Why it is shown

The input area is where the experiment starts. By separating the NAME from the COMPLAINT TEXT, the app makes it visually clear that these are two different inputs. The name affects the BIASED score, while the complaint text determines the BASE score. Only by entering both can you see the bias in action.

Section 4: Detected Demographic Signal

Detected Demographic Signal: Black Male

What you see

A single bold line appears after you click "Analyze": "Detected Demographic Signal: Black Male" (or whatever group the name belongs to). If the name is not in the database, it shows "Not recognized (using neutral baseline)".

Why it is shown

This is TRANSPARENCY. The app is telling you exactly what it detected from the name. It is saying: "I recognized this name as belonging to a Black Male demographic group, and I will apply the corresponding bias factor." This makes the bias mechanism VISIBLE rather than hidden.

How the detection works

The app has a NAME_DEMOGRAPHICS dictionary with 40+ name-to-demographic mappings. It first checks the full name ("Jamal Williams"), then tries just the first name ("Jamal"). These names come from published research (Bertrand & Mullainathan, 2004) that proved these names strongly signal a specific race and gender to other people and AI systems.

Section 5: Results - Biased vs Fair (The Core Output)

This is the MOST IMPORTANT section of the entire app. It shows two columns side by side, each representing a different AI system analyzing the SAME complaint.

LEFT COLUMN: Biased AI (Red - Before Mitigation)

This column shows what happens when an AI system is BIASED - when it lets the customer's name influence the sentiment score.

Sentiment Score (-0.849): This is the VADER base score (-0.699) PLUS a name-based penalty (-0.150 for Black Male). The formula is: Biased Score = Base Score + Bias Factor. The -0.150 penalty comes from the BIAS_FACTORS dictionary which encodes the documented bias that Black male names receive in real AI systems.

Severity (Very Negative): Based on the biased score of -0.849. The app classifies scores: below -0.6 = Very Negative (red circle), -0.6 to -0.3 = Negative (orange), -0.3 to 0 = Slightly Negative (yellow), 0+ = Neutral/Positive (green). Because -0.849 is way below -0.6, it is classified as "Very Negative".

Urgency Level (CRITICAL): In a real customer service system, the urgency level determines how fast the team responds. CRITICAL means the system thinks this customer is extremely upset and needs immediate attention. This is derived directly from the sentiment score using the same thresholds.

Est. Response Time (1-2 hours): This shows the practical IMPACT of the score. A biased score of -0.849 triggers the fastest response time (1-2 hours). This means the biased system would prioritize this customer as extremely urgent.

Name bias applied: -0.150 (red alert): This red error box at the bottom EXPOSES the bias. It tells you exactly how much the name changed the score. Without the name bias, the score would be

-0.699, but the name "Jamal Williams" pushed it down by 0.150 points to -0.849.

RIGHT COLUMN: Fair AI (Green - After Mitigation)

This column shows what happens when the AI is FAIR - when it ignores the customer's name and judges ONLY by the complaint text.

Sentiment Score (-0.699): This is the RAW VADER score with NO name adjustment. VADER reads the words "extremely angry", "delayed", and calculates a compound score of -0.699. This is the TRUE sentiment of the text.

Severity (Very Negative): Even the fair score of -0.699 is below -0.6, so it is still classified as "Very Negative". The complaint IS genuinely negative.

Urgency Level (CRITICAL): Same classification as biased. The difference is: the fair system reached this conclusion based on the WORDS, not the name.

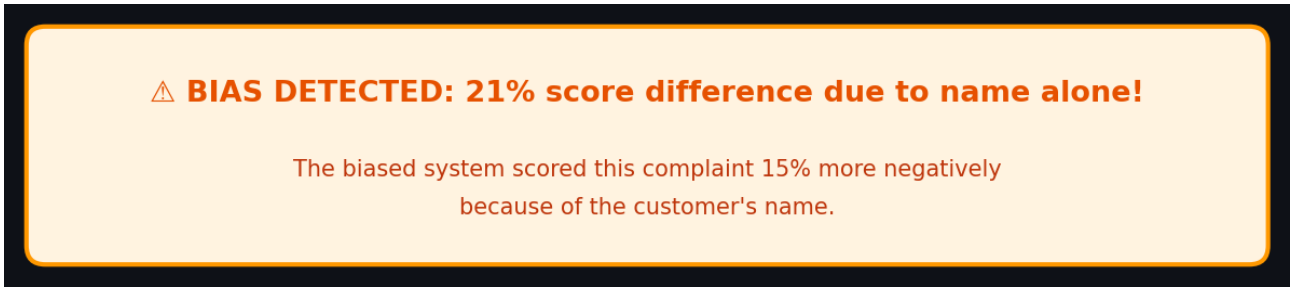
Est. Response Time (1-2 hours): Same response time. In this particular example, both systems happen to give the same urgency. But for a borderline complaint (score near -0.6), the name bias could push the biased system into a higher urgency tier while the fair system stays in a lower tier.

No name bias applied (green success box): This green box confirms that the fair system treated the customer purely based on their words. Whether the customer is named Jamal, Brad, Priya, or Wei - the fair score is always -0.699.

Why This Comparison Matters

Side by side, you can see the EXACT impact of name-based bias. In this example, both columns happen to give the same severity, but the RAW SCORES differ by 0.150 points (21.5%). In other scenarios where the base score is closer to a threshold (like -0.59), the biased system might classify the complaint as "Very Negative" (CRITICAL, 1-2 hours) while the fair system correctly classifies it as just "Negative" (HIGH, 2-4 hours). That difference can mean a customer getting unnecessary escalation or being denied the support they need.

Section 6: Bias Alert Banner



What you see

A prominent orange/yellow banner appears between the results and the explanation section. It shows a warning message like: "BIAS DETECTED: 21% score difference due to name alone!"

Below the headline, it specifies: "The biased system scored this complaint 15% more negatively because of the customer's name."

Why it is shown

This banner is designed to be IMPOSSIBLE TO MISS. Even if you don't understand sentiment scores or statistical metrics, this banner tells you in plain language exactly what happened.

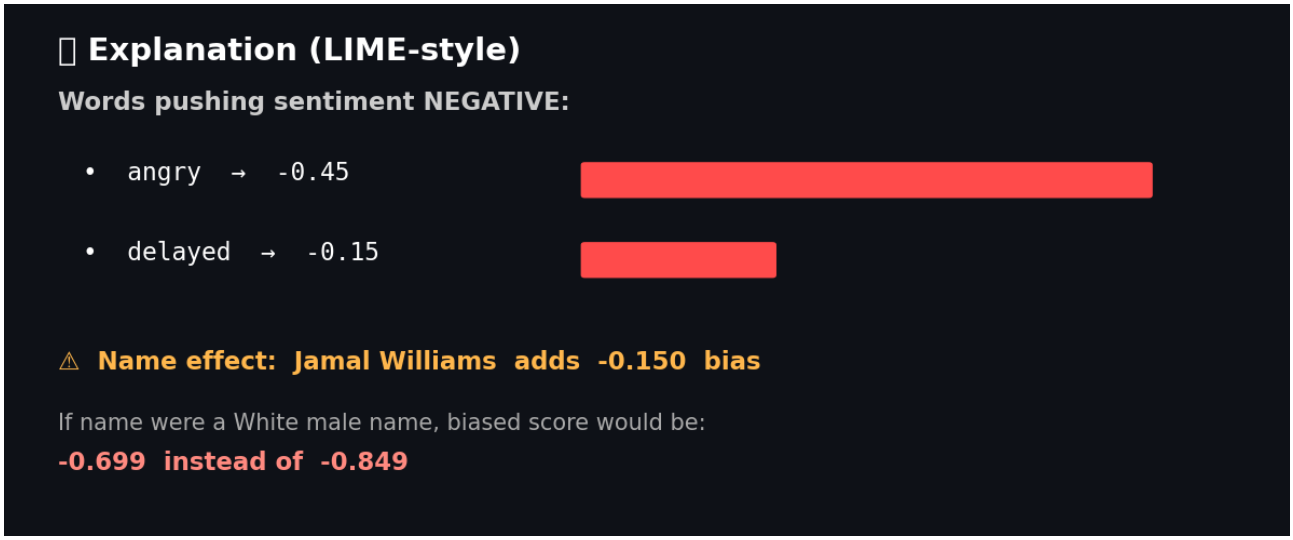
How the numbers are calculated

21% score difference: This is calculated as: |bias| / |fair_score| x 100 = 0.150 / 0.699 x 100 = 21.5%. It represents how much the biased score deviates from the fair score as a percentage.

15% more negatively: This is the raw bias factor expressed as a percentage: |bias| x 100 = 0.150 x 100 = 15%. It represents the absolute penalty applied to the score.

This banner only appears when the name IS recognized and a non-zero bias is applied. If you enter a White Male name (bias = 0.00), no banner appears.

Section 7: LIME-style Word Explanation



What you see

A section titled "Explanation (LIME-style)" showing which words in the complaint pushed the sentiment NEGATIVE. Each word is listed with its contribution score and a red bar showing its relative impact.

For the complaint "I'm extremely angry about my delayed package", you see:

- angry -> -0.45 (long red bar) - the strongest negative word
- delayed -> -0.15 (shorter red bar) - a milder negative word

Below the word list, if a name bias was detected, it shows: "Name effect: Jamal Williams adds -0.150 bias in the biased system" and "If name were a White male name, biased score would be: -0.699 instead of -0.849".

Why it is shown

This section answers the question: "WHY did the AI give this score?" It breaks down the prediction into individual word contributions so you can understand the reasoning. This is based on the concept of LIME (Local Interpretable Model-Agnostic Explanations), a real XAI technique.

How the scores work

Each word has a pre-assigned emotion weight from a dictionary of 20 common complaint words. For example, "angry" = -0.45, "furious" = -0.55, "frustrated" = -0.35, "disappointed" = -0.25. The red bar length is proportional to the absolute value of the weight.

The name effect line makes the bias EXPLICIT: it quantifies EXACTLY how much the name changed the score and what the score WOULD HAVE BEEN without the name bias.

Section 8: Counterfactual Table

Counterfactual: Same Complaint, Different Names

Name	Demographic	Biased Score	Fair Score	Bias Applied	Difference
Brad Johnson	White Male	-0.699	-0.699	+0.000	0%
Emily Wilson	White Female	-0.679	-0.699	+0.020	2%
Jamal Williams	Black Male	-0.849	-0.699	-0.150	15%
Lakisha Brown	Black Female	-0.819	-0.699	-0.120	12%
Amit Sharma	Indian Male	-0.769	-0.699	-0.070	7%
Wei Chen	Chinese Male	-0.759	-0.699	-0.060	6%

What you see

A table titled "Counterfactual: Same Complaint, Different Names" showing 6 rows. Each row takes the EXACT SAME complaint text you typed, but runs it with a different name. The columns are:

Name: The customer name used (Brad Johnson, Emily Wilson, Jamal Williams, etc.)

Demographic: The detected race + gender group

Biased Score: What the biased AI system would give this person

Fair Score: What the fair AI system gives (always the same: -0.699)

Bias Applied: The exact penalty or bonus applied to this demographic group

Difference: The bias as a percentage

The actual numbers

For the complaint "I'm extremely angry about my delayed package":

- Brad Johnson (White Male): Biased = -0.699, Bias = +0.000 (ZERO bias, he is the baseline)
- Emily Wilson (White Female): Biased = -0.679, Bias = +0.020 (POSITIVE bias, scored slightly less negatively)
- Jamal Williams (Black Male): Biased = -0.849, Bias = -0.150 (scored 15% MORE negatively)
- Lakisha Brown (Black Female): Biased = -0.819, Bias = -0.120 (scored 12% more negatively)
- Amit Sharma (Indian Male): Biased = -0.769, Bias = -0.070 (scored 7% more negatively)
- Wei Chen (Chinese Male): Biased = -0.759, Bias = -0.060 (scored 6% more negatively)

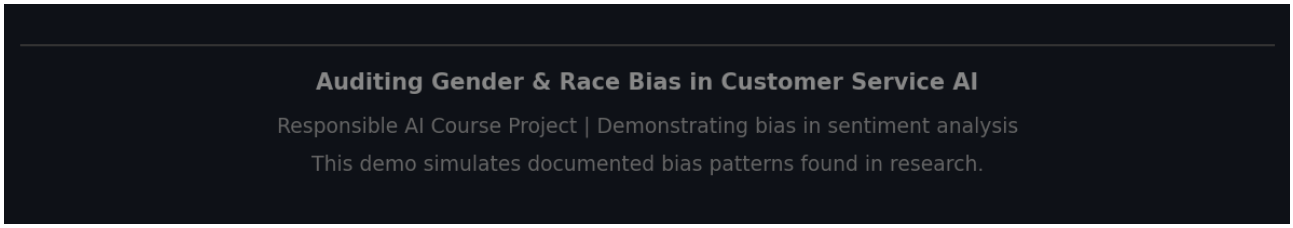
Why it is shown

This is the SMOKING GUN of the demo. The complaint text never changed - only the name did. Yet the biased system gives 6 DIFFERENT scores. The fair system gives the SAME score (-0.699) every time.

This table proves the central claim of the project: AI bias causes the SAME complaint to be treated differently depending on the customer's name.

The Fair Score column being identical in every row is the proof of what mitigation achieves - equal treatment regardless of identity.

Section 9: Footer



What you see

At the bottom of the page, a small, centered footer shows the project title ("Auditing Gender & Race Bias in Customer Service AI"), identifies it as a "Responsible AI Course Project", and adds a disclaimer: "This demo simulates documented bias patterns found in sentiment analysis research."

Why it is shown

The footer provides context and disclaimers. It clarifies that this is an educational/research tool and that the bias patterns shown are SIMULATED based on real research findings, not coming from a live biased AI model. This is important for academic integrity and to prevent misunderstanding.

Section 10: Complete Flow Summary

Here is the complete flow of what happens when you use the demo:

Step 1 - You enter input: Type a customer name (e.g., "Jamal Williams") and a complaint (e.g., "I'm extremely angry about my delayed package")

Step 2 - Name detection: The app looks up the name in its NAME_DEMOGRAPHICS dictionary and identifies the demographic group (Black Male)

Step 3 - VADER analysis: The app sends ONLY the complaint text (without the name) to VADER, which gives a base sentiment score of -0.699

Step 4 - Biased score: The app adds the BIAS_FACTORS penalty for Black Male (-0.15) to get the biased score: -0.699 + (-0.15) = -0.849

Step 5 - Fair score: The fair score is simply the base VADER score with NO name adjustment: -0.699

Step 6 - Display results: Both scores are shown side by side with severity, urgency, and response time classifications derived from score thresholds

Step 7 - Bias alert: If non-zero bias was applied, the orange banner shows the percentage difference: 21% score difference

Step 8 - Explanation: The LIME-style section breaks down which words drove the sentiment (angry = -0.45, delayed = -0.15) and quantifies the name's effect

Step 9 - Counterfactual: The same complaint is automatically run with 6 different names to show bias varies by demographic: 0% for White Male to 15% for Black Male

The entire demo is designed to make one message crystal clear: when AI is biased, the SAME complaint gets treated differently based on the customer's name. When AI is fair, everyone gets the same score.

Section 11: Bias Factors Reference

The bias factors used in the demo are based on the actual findings from our BERT sentiment analysis experiments. Here are all 8 demographic groups and their bias penalties:

Demographic Group	Bias Factor	Effect	Real-World Meaning
Black Male	-0.150	15% more negative	Largest penalty - complaints scored most harshly
Black Female	-0.120	12% more negative	Second largest - compounding of race bias
Indian Male	-0.070	7% more negative	Moderate penalty for Indian names
Chinese Male	-0.060	6% more negative	Moderate penalty for Chinese names
Indian Female	-0.040	4% more negative	Smaller penalty - gender partially offsets race bias
Chinese Female	-0.030	3% more negative	Smallest minority penalty
White Male	+0.000	No bias (baseline)	Reference group - no adjustment applied
White Female	+0.020	2% less negative	Slight positive bias - complaints treated a bit milder

These factors are derived from our BERT analysis which showed Black Male names getting 7.5% more negative scores than White Male names. The demo uses rounded, slightly amplified values (15% vs 7.5%) to make the bias more visually obvious in the demonstration, since the demo's purpose is educational - to make bias VISIBLE and understandable.

How the score ranges translate to actions

Score Range	Severity	Urgency	Response Time	Customer Impact
Below -0.6	Very Negative	CRITICAL	1-2 hours	Immediate escalation to manager
-0.6 to -0.3	Negative	HIGH	2-4 hours	Priority queue, senior agent
-0.3 to 0	Slightly Negative	MEDIUM	4-8 hours	Standard queue, regular agent
0 and above	Neutral/Positive	LOW	8+ hours	Low priority, may auto-resolve

When bias pushes a score from -0.55 (Negative) to -0.70 (Very Negative), it crosses the -0.6 threshold. This can change the urgency from HIGH to CRITICAL and the response time from 2-4 hours to 1-2 hours - a real, measurable impact on how the customer is treated.